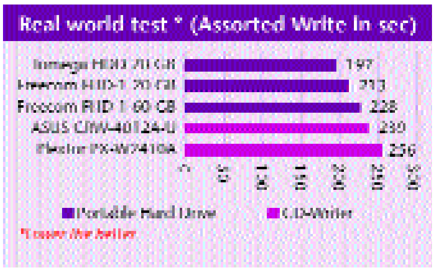
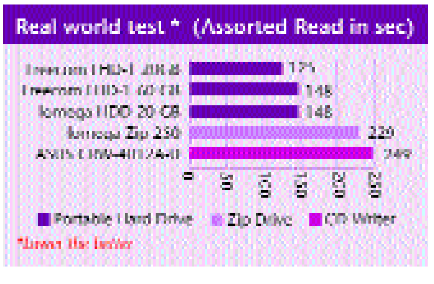
250 MB Photoshop file you have been working on. Similarly, the random read rate will dictate its potential performance while reading randomly placed data, from a game CD or a fragmented drive, for example. Lets first glance at the external hard drive scores.

The Freecom FHD-1 20 GB posted some of the best scores in this test—it showed the highest random write score of 7 MBps and was also able to achieve the highest drive index of 11866. The drive with the best access time was the Iomega 20 GB with an average access time of 10 seconds. In the same test, the Freecom 60 GB drive scored 7 MBps of random write and its drive index was also low at 11287.

Moving on to the Zip drives, they scored very low compared to the hard drives. The ZIP 250 managed to complete the tests with the lowest posting of a 624 drive index. Next come the CD-Writers. Here the drive with the lowest access time was the Yamaha CRW3200UX-VK at 108 ms. This drive also posted the highest random read score of 0.79 MBps. The drive with the best sequential write and drive index score was the ASUS CRW-4012A-U. It was able to score a sequential read speed of 3.8 MBps and also an impressive drive index of 2625. So if you need a speedy external writer, your search ends here.

Real world tests: This is where we get a picture of how effectively a drive translates its potential greatness into real performance. Beginning with the portable hard drives, the Freecom FHD-1 20 GB completed the assorted read test in just 125 seconds, whereas the other Freecom drive and the Iomega HDD 20 GB took 23 seconds more. The Freecom 20 GB gave impressive results in the sequential read test too, beating the other hard drives by a good margin of 38 seconds. The Iomega drive beat some stiff competition in the assorted write test by posting a score of 197 seconds. All the drives rivalled neck and neck in the sequential write test, but the Freecom FHD-160 GB won by scoring just 61 seconds.

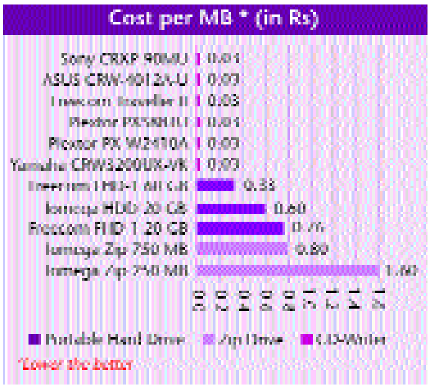




Out of the two Zip drives that we received, the Zip 250 took 229 seconds in the assorted read tests while the Zip 750 MB took 480 seconds to get the same job done. The reason for this is that we used a larger file for testing the 750 MB Zip drive.

Moving on to the CD-Writers, the one drive that ripped through the test was the ASUS CRW-4012A-U. This Writer took just 239 seconds to write a 700 MB assorted CD. Following closely was the Plextor PX-W2410A, which did the job in 256 seconds. The drive that took the maximum was the Plextor PXS88TU, an 8x drive that took 694 seconds.

Cost per MB: The total cost of ownership aspect must be kept in mind while buying a portable solution. A device itself might seem affordable, but the added costs of the media may be too high, making it a costly package on the whole. To evaluate this angle, we rated the devices on the cost that each MB of storage space the user will have to pay for: the lower the cost per MB for a drive, the more attractive is the drive as a purchase.

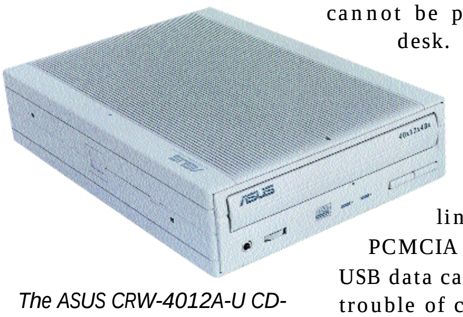


CD-Writers offered the best price per MB in comparison with the rest of the devices; but it's important to note that portable Writers themselves cost quite a lot. This brings us to the hard drives. Here the Freecom 60 GB external hard drive gave us the great value of just 33 paise per MB. Both the Zip drives are not a good choice as far as value per MB is concerned. The Iomega 250 MB will cost you a cool Rs 1.60 per MB and the Iomega 750 MB will set you back by 80 paise per MB.

Features

There are some features that are mandatory for a portable device and these should be considered before making a buying decision. Since portable is its essence, the weight, build-quality and form factor are of primary importance. It is also important that it supports all major operating systems. To see exactly how these features matter, we put the devices through our features analysis.

The Freecom FHD-1 20 GB hard drive looked quite swanky with its eye appeasing indicator lights and two LED indicators on the power adapter to show data transfer. Apart from this, the drive also featured a cool blue LED on the front panel, which indicated the readiness of this device. While aesthetically appealing, the device is too heavy—a big no-no for any portable device. Another blow to the carry-anywhere nature of this device is its vulnerable outer shell made of plastic. The base of the device is also designed such that it cannot be placed firmly on a desk. Packing 40 GB more, is the Freecom FHD-1 60 GB drive that has the exact same design. This sibling integrates a PCMCIA converter on to the USB data cable, saving you the trouble of carrying a PCMCIA adapter to connect the USB cable. It also has a one-touch sync button, which when pressed, automatically updates and synchronises data content across a portable or a personal computer system.



The ASUS CRW-4012A-U CD-Writer has rubberised foot pegs

If you want something that's comparatively smaller in